

Lecture 10 – Wired LANs: Ethernet (ELC-411)

Q1: What is a Local Area Network (LAN)?

A Local Area Network (LAN) is a computer network designed to cover a small geographic area such as a building or campus. LANs allow devices to share resources and may connect to WANs or the Internet.

Q2: What is IEEE Project 802?

IEEE Project 802 defines standards for Local Area Networks, covering the Physical and Data Link layers to ensure interoperability between devices from different manufacturers.

Q3: Explain the IEEE 802 Data Link Layer structure.

The Data Link Layer is divided into Logical Link Control (LLC), which provides flow and error control, and Media Access Control (MAC), which defines access methods and framing.

Q4: What is Ethernet? Why is it popular?

Ethernet is a LAN technology standardized under IEEE 802.3. It is popular due to its simplicity, low cost, scalability, and support for high data rates.

Q5: Explain the evolution of Ethernet.

Ethernet evolved from Standard Ethernet (10 Mbps) to Fast Ethernet (100 Mbps), Gigabit Ethernet (1 Gbps), and Ten-Gigabit Ethernet (10 Gbps).

Q6: Explain the Ethernet MAC frame format.

The Ethernet frame includes Preamble, SFD, Destination Address, Source Address, Length/Type field, Data and Padding, and CRC for error detection.

Q7: What are the minimum and maximum Ethernet frame sizes? Why is the minimum size important?

The minimum frame size is 64 bytes and the maximum is 1518 bytes. The minimum size ensures collision detection in CSMA/CD networks.

Q8: Explain Ethernet addressing and its types.

Ethernet uses a 48-bit MAC address. Address types include Unicast, Multicast, and Broadcast.

Q9: How can you distinguish between unicast and multicast Ethernet addresses?

The least significant bit of the first byte determines the type: 0 for unicast and 1 for multicast. The broadcast address has all bits set to 1.

Q10: Explain 10Base5 Ethernet.

10Base5 uses thick coaxial cable, bus topology, supports 10 Mbps, and has a maximum segment length of 500 meters.

Q11: Explain 10Base2 Ethernet.

10Base2 uses thin coaxial cable with bus topology, supports 10 Mbps, and has a maximum length of 185 meters.

Q12: Explain 10Base-T Ethernet.

10Base-T uses twisted pair cables in a star topology with hubs and supports a maximum length of 100 meters.

Q13: Explain the role of bridges in Ethernet.

Bridges divide a LAN into segments, reduce collisions, and increase overall bandwidth.

Q14: What is Switched Ethernet?

Switched Ethernet uses switches instead of hubs, giving each station a dedicated collision domain.

Q15: Explain Full-Duplex Ethernet.

Full-Duplex Ethernet allows simultaneous transmission and reception, eliminating collisions and doubling effective bandwidth.

Q16: Explain Fast Ethernet.

Fast Ethernet operates at 100 Mbps, is standardized under IEEE 802.3u, and is backward compatible with Standard Ethernet.

Q17: Explain Gigabit Ethernet.

Gigabit Ethernet operates at 1000 Mbps under IEEE 802.3z and supports full-duplex communication.